CRITERION NETWORKS

PDFModelContextProtocol.pdf

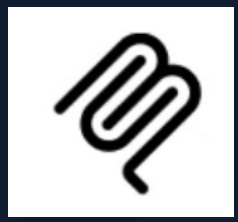
CONFIDENTIAL
https://www.criterionnetworks.com/

Criterion Networks Document

Copyright © 2021 Criterion Networks Inc. All Rights Reserved. 2025-10-22 14:28:1761143282
Protected with free version of Make Watermark. Full version doesn't put this mark.

CONFIDENTIAL
https://www.criterionnetworks.com/

Model Context Protocol (MCP) Essentials for AI-Driven Network Operations



Criterion

CONFIDENTIAL
https://www.criternetworks.com/

About Me



Sharad Chandra

Principal Solution Architect (Criterion Networks)

Key Expertize

- Data Networks, Cloud, SDN
- Information Security
- Observability - Thousand Eyes, Splunk
- AI Generalist and Evangelist
- Emeritus CCIE #19273 (Security), CCDP

Experience

- Work Experience (since 2003)
- Criterion Networks
- Aryaka Networks
- Cisco Systems
- British Telecom
- Dimension Data

Contact Details

- Email: schandra@criterionnetworks.com
- LinkedIn: https://www.linkedin.com/in/chandrasharad/

CONFIDENTIAL https://www.criterionnetworks.com/

Criterion Proprietary & Confidential

Criterion

Agenda

1. **Understanding LLMs & AI Applications**
   Large Language Models, AI-powered applications, and agentic workflows
2. **The Integration Challenge**
   API architecture, the MxN problem, and current limitations
3. **Introducing Model Context Protocol**
   MCP architecture, capabilities, and the paradigm shift
4. **Technical Implementation**
   Core capabilities, security, SDKs, and protocol extensions
5. **Getting Started & Future Vision**
   Implementation guide, resources, and the future of AI integration

CONFIDENTIAL
https://www.criterionnetworks.com/

Section 1

Understanding Large Language Models (LLMs)

CONFIDENTIAL
https://www.criterionnetworks.com/

Criterion

What Are Large Language Models?

Definition

Large language models (LLMs) are deep learning models trained on immense amounts of data, capable of understanding and generating natural language to perform a wide range of tasks.

Core Mechanism

LLMs work as giant statistical prediction machines that repeatedly predict the next word in a sequence. They learn patterns in text and generate language that follows those patterns.

Breakthrough Technology

LLMs represent the first AI system that can handle unstructured human language at scale, allowing for natural communication with machines.

CONFIDENTIAL
https://www.criterionnetworks.com/

How LLMs Work: The Transformer Architecture

Built on neural networks called transformers, introduced in 2017

Key Components & Training Process

Tokenization & Embeddings

Text is broken into tokens (words/subwords) and mapped to numerical vectors called embeddings that capture semantic meaning.

Self-Attention Mechanism

Each token "pays attention" to other tokens to understand context. The model calculates relationships between all tokens, determining which words matter most to each other.

Parameters & Training

LLMs contain billions of parameters (weights) that are optimized through training on massive datasets. The model learns patterns by predicting the next word millions of times.

CONFIDENTIAL
https://www.criterionnetworks.com/

Criterion

AI Applications Powered by LLMs

Conversational AI & Chatbots

Virtual assistants like ChatGPT, Claude, and Gemini provide natural, human-like interactions for customer service, support, and general assistance.

Enterprise Document Analysis

Automated review of legal contracts, medical records, and business documents for compliance, insights, and decision support.

Content Creation & Summarization

Automated generation of marketing copy, articles, reports, and condensing long documents into key insights for business intelligence.

Translation & Multilingual Support

Real-time language translation and localization enabling global communication and content adaptation across markets.

Code Generation & Development

AI-powered coding assistants write, debug, and optimize software code, accelerating development cycles and reducing errors.

Sentiment & Classification

Analyzing customer feedback, social media, and communications to detect sentiment, toxicity, and categorize content automatically.

CONFIDENTIAL
https://www.criterionnetworks.com/

AI Agent vs Agentic Workflow

Aspect	AI Agent	Agentic Workflow
Definition	Single autonomous entity designed to handle specific tasks within defined parameters	Multi-step process orchestrating multiple AI agents, services, and APIs dynamically
Architecture	Opaque (black box) with minimal external control and internal reasoning loop	Modular, traceable with externalized control flow and decision nodes
Autonomy	Operates independently within designated domain; decisions internalized	Coordinates multiple systems; decisions externalized to workflow nodes
Complexity	Handles specific tasks with clear patterns; improves through updates	Manages complex multi-step processes; learns from interactions and adapts
Use Cases	Chatbots, scheduling assistants, code generation, customer service	Enterprise IT support, fraud detection, research coordination, threat analysis
Governance	Limited observability and control once deployed	Built-in validation, human oversight, and observability features

CONFIDENTIAL
https://www.criterionnetworks.com/

Criterion

Agentic Workflow

Interpret goals from user input or system instructions

- e.g., "Book me a flight to Rome next Friday"

Break down tasks into sub-tasks, often using a planning module.

Take actions by calling tools or APIs, accessing databases, or interacting with external environments (such as browsing the web or sending an email).

Evaluate outcomes of its actions and adapt its next steps accordingly.

CONFIDENTIAL
https://www.criterionnetworks.com/

Agentic AI Evolution: A Journey

1 Early Foundations (1950s 1980s)

From the 1950 Turing Test through the 1956 Logic Theorist and 1966 ELIZA chatbot, early AI demonstrated that machines could engage in structured problem-solving and basic interaction. These systems split into symbolic AI (using logical rules and knowledge representation) and statistical AI (focusing on probability and pattern recognition), but neither could learn from experience or handle unexpected situations independently.

2 The Paradigm Shift to Autonomous Intelligence

Unlike traditional AI that waits for human direction, agentic AI initiates actions, prioritizes tasks, and adapts autonomously through self-directed, context-aware decision-making. These systems fuse reasoning with goal-driven autonomy, capable of contextual assessment, navigating uncertainty, and taking independent action to orchestrate workflows and resolve issues across complex value chains.

3 Core Architectural Components

AI agents integrate five essential components: language models as the reasoning "brain," memory for context storage, tools for external system interfaces, planning modules for goal-setting and action selection, and reflection loops for continuous improvement. This architecture supports complex reasoning, planning, and adaptive behavior that combines reactive responses with goal-based planning and learned strategies that evolve over time.

CONFIDENTIAL
https://www.criterionnetworks.com/

Criterion

Agentic AI **Transforms** Enterprise Operations

Industry Applications at Scale

Agents autonomously manage manufacturing variants, optimize supply chains in real-time, coordinate clinical workflows in healthcare, and orchestrate compliance across financial systems.

They continuously adapt to changing conditions, reducing human intervention while maintaining operational excellence.

Critical Challenges & Governance

Enterprise transformation requires addressing trust, traceability, and data readiness. Standardized authentication patterns ensure secure access, while consistent message formats make all agent-system interactions auditable and explainable.

This creates sustainable architecture with transparent bounds on agent behavior.

The Future: Hybrid Human-AI Ecosystems

AI agents will assume persistent roles across product lifecycles, autonomously discovering and using tools, gathering real-time context for multi-step reasoning (supporting CoT and ToT frameworks), and orchestrating workflows across complex value chains.

This enables the shift from reactive to proactive AI—faster, more adaptive, and infinitely more scalable.

CONFIDENTIAL
https://www.criterionnetworks.com/

How APIs Work: Technical Architecture

Request-Response Cycle

- Client sends HTTP request to server endpoint
- Server processes request and returns HTTP response
- Response includes status code, headers, and data payload

HTTP Methods & Endpoints

- GET: Retrieve data from server
- POST: Create new resources
- PUT/PATCH: Update existing resources
- DELETE: Remove resources

Request Components

- Headers: Metadata (auth tokens, content type)
- Parameters: Query strings or path variables
- Body: Data payload (JSON, XML, form data)

Data Formats & Protocols

- JSON: Lightweight, human-readable format
- REST: Architectural style using HTTP methods
- GraphQL: Query language for flexible data retrieval

Stateless Architecture

Each API request is independent and contains all necessary information. The server doesn't maintain session state between requests.

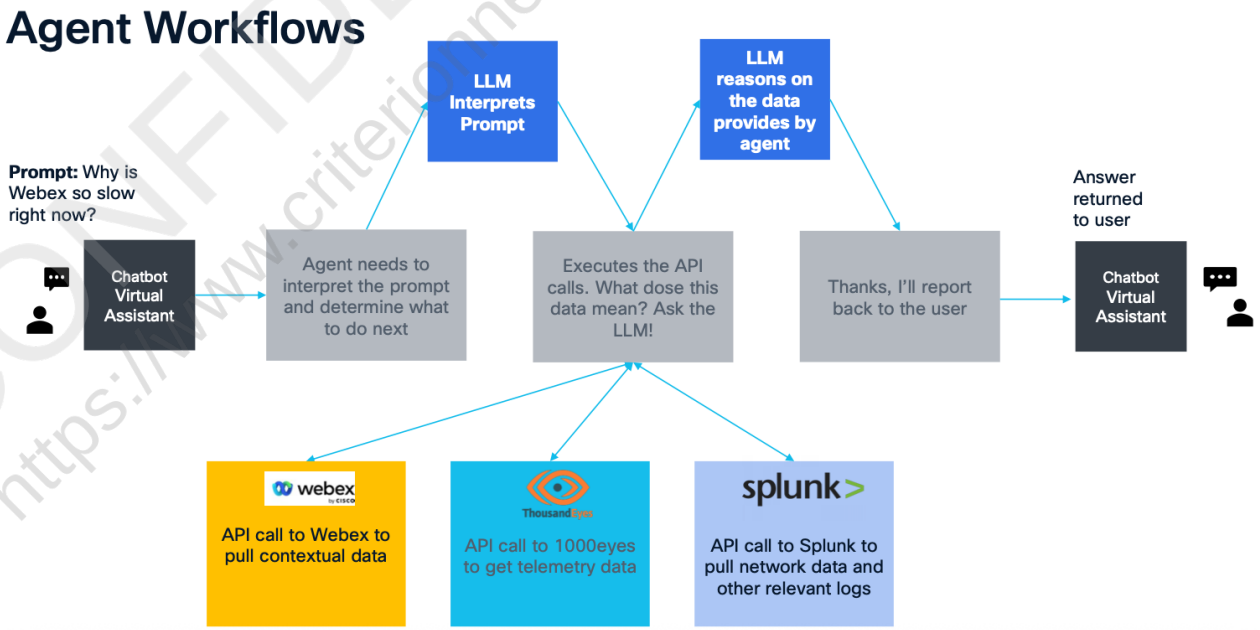
Benefits:

- Scalability: Servers can handle requests without memory overhead
- Reliability: No session data loss if server fails
- Simplicity: Easier to cache and distribute requests

Authentication: Typically handled via tokens (JWT, OAuth) passed in headers rather than server-side sessions.

CONFIDENTIAL
https://www.criterionnetworks.com/

The MxN Integration Problem



The Challenge: Each AI agent requires custom integration code for every external system. With M agents and N systems, this creates M×N integration points that must be built, tested, and maintained separately.

CONFIDENTIAL
https://www.criterionnetworks.com/

Criterion

The Integration Challenge for AI Tools

Key Challenges

The N×M Problem

Connecting N AI applications to M data sources requires building N×M custom integrations. This exponential complexity creates massive technical debt and slows innovation.

Current Pain Points

- Custom code required for each integration (150+ lines per tool)
- Brittle integrations that break with API changes
- Constant maintenance overhead across all connections
- Limited tool discovery and dynamic capability detection

Industry Impact

Organizations spend 60-70% of development time on integration work rather than core AI capabilities. This creates vendor lock-in, siloed solutions, and high barriers to entry for AI adoption.

CONFIDENTIAL
https://www.criterionnetworks.com/

APIs & The MxN Problem

Connecting **M** AI applications to **N** data sources requires building **M×N** custom integrations.

Example:

5 AI apps × 10 data sources = **50 integrations**

This exponential complexity creates massive development overhead, maintenance burden, and integration fragility.

The MCP Solution: M+N

Linear Scaling

MCP reduces complexity from **M×N** to **M+N** by introducing a standardized protocol layer.

Example:

5 AI apps + 10 MCP servers = **15 integrations**

Key Benefits

- **67% reduction** in integration effort (50 → 15)
- **Unified authentication** and error handling
- **Standardized interfaces** across all tools
- **Dynamic discovery** of new capabilities

CONFIDENTIAL
https://www.criterionnetworks.com/

Criterion

What is **Model Context Protocol** and why does it **matter**?



Definition	Standardization protocol for AI tool integration "USB-C for AI" - universal connector NOT an AI assistant or agent
Core Purpose	Establish client-server architecture Connect AI applications to external tools Enable dynamic tool discovery
Key Benefits	Linear scaling (M+N) vs exponential (M×N) Unified authentication and error handling Standardized interfaces across tools

CONFIDENTIAL
https://www.criterionnetworks.com/

Criterion

MCP
Core Capabilities

Standardized protocol primitives for AI-tool integration

Tools	Resources	Prompts
● Function calling and execution	● Contextual data access	● Template management
● Dynamic tool discovery	● File and database integration	● Reusable prompt libraries
● Structured input/output schemas	● Real-time resource updates	● Parameterized interactions
● Multi-modal result types	● Subscription-based notifications	● LLM-agnostic design

CONFIDENTIAL
https://www.criterionnetworks.com/

CONFIDENTIAL
http://www.criterionnetworks.com/

MCP Enables **Scalable, Interoperable** Agent Ecosystems

Integration Complexity & Deployment

By providing a universal interface, MCP eliminates the need to build and maintain separate connectors for each data source. Pre-built MCP servers are available for popular enterprise systems including Google Drive, Slack, GitHub, Postgres, and development tools like Zed, Replit, and Sourcegraph.

Organizations build each integration once as an MCP server, and any AI agent that speaks MCP can immediately leverage it, dramatically reducing time from concept to production while simplifying ongoing maintenance.

Autonomy & Governance

MCP addresses the critical challenges of trust, traceability, and data readiness identified in enterprise transformation. The standardized authentication patterns (OAuth, bearer tokens, API keys) ensure secure access, while the consistent message format makes all agent-system interactions auditable and explainable.

This creates the sustainable architecture needed for agents to independently interface with PLM, ERP, digital twins, and legacy systems while maintaining transparent bounds on agent behavior.

MCP Enables **Scalable, Interoperable** Agent Ecosystems

(continued)

Infrastructure for Hybrid Human-AI Ecosystems

As organizations assign AI agents persistent roles across product lifecycles—managing variants, orchestrating compliance, and continuously optimizing operations—MCP provides the foundational connectivity that makes this possible.

The protocol supports the shift from reactive to proactive AI by enabling agents to autonomously discover and use tools, gather real-time context for multi-step reasoning (supporting CoT and ToT frameworks), and orchestrate workflows across complex value chains. This positions MCP as the essential enabling technology for the future of machine-empowered humanity that is faster, more adaptive, and infinitely more scalable.

CONFIDENTIAL
https://www.criterionnetworks.com/

The MCP Paradigm Shift

A fundamental transformation in AI tool integration

	Before MCP	With MCP
Development	• Custom coding for each integration • 150+ lines of code per tool • Multiple authentication systems	• Standardized protocol implementation • 5 lines of code for connection • Unified authentication
Maintenance	• Exponential complexity (N×M) • Brittle integrations • Constant updates required	• Linear scalability (M+N) • Resilient connections • 90% less maintenance overhead
Innovation	• Siloed solutions • Vendor lock-in • Limited tool discovery	• Interoperable ecosystems • Vendor independence • Dynamic tool discovery
Adoption	• Technical expertise required • High barrier to entry • Limited accessibility	• Accessible AI development • Democratized integration • Broader participation

Criterion

CONFIDENTIAL
https://www.criterionnetworks.com/

CONFIDENTIAL
https://www.criterionnetworks.com/

Technical Deep Dive

Model Context Protocol

A standardized protocol for connecting AI applications to external systems, built on JSON-RPC for seamless integration of tools, resources, and prompts.

Protocol Version: 2025-06-18

Criterion

CONFIDENTIAL
https://www.criterionnetworks.com/

MCP Protocol Architecture

Technical components and protocol layers

Core Components

Host

The user-facing AI application that orchestrates interactions between users, LLMs, and external tools.

Examples: Claude Desktop, Cursor IDE

Client

A component within the Host that maintains a 1:1 connection with a Server, handling MCP communication details.

Server

An external program that exposes capabilities (Tools, Resources, Prompts) via the MCP protocol. Can run locally or remotely.

Protocol Layers

Data Layer

Defines the JSON-RPC based protocol for client-server communication:

- **Lifecycle:** Connection initialization and termination
- **Primitives:** Core data structures and messages
- **Notifications:** Asynchronous event updates

Transport Layer

Defines the communication mechanisms between clients and servers:

- **Stdio:** Local process communication via standard I/O
- **HTTP+SSE:** Remote communication with Server-Sent Events
- **Auth:** Authentication and authorization mechanisms

Protocol Version: 2025-06-18

JSON-RPC: The Technical Foundation

MCP's communication protocol implementation

1. Requests

Sent from Client to Server to initiate an operation. Includes an ID, method name, and parameters.

```
{
"jsonrpc": "2.0",
"id": 1,
"method": "tools/call",
"params": {
"name": "weather",
"arguments": {
"location": "San Francisco"
}
}
}
```

2. Responses

Sent from Server to Client in reply to a Request. Includes the same ID and either a result or an error.

```
{
"jsonrpc": "2.0",
"id": 1,
"result": {
"temperature": 62,
"conditions": "Partly cloudy"
}
}
```

CONFIDENTIAL
https://www.criterionnetworks.com/

JSON-RPC: The Technical Foundation

MCP's communication protocol implementation

(continued)

3. Notifications

One-way messages that don't require a response. Used for updates or events.

```
{
"jsonrpc": "2.0",
"method": "notifications/tools/list_changed"
}
```

Protocol Version: 2025-06-18

CONFIDENTIAL
https://www.criterionnetworks.com/

MCP Transport Mechanisms

Technical implementation of communication channels

Stdio Transport

Local communication where Client and Server run on the same machine via standard input/output streams.

Implementation

Server launched as subprocess by Host. Client writes to stdin and reads from stdout.

Technical Benefits

- No network config
- Low latency

Streamable HTTP Transport

Remote communication where Client and Server might be on different machines using HTTP POST with Server-Sent Events.

Implementation

Client sends requests via HTTP POST. Server uses SSE for streaming responses and notifications.

Authentication Methods

- Bearer tokens
- API keys

CONFIDENTIAL
https://www.criterionnetworks.com/

MCP Protocol Lifecycle

Technical sequence of protocol interactions

1. Initialization

Client connects to Server and they exchange protocol versions and capabilities.

```
// Client → Server
{ "jsonrpc": "2.0", "id": 1, "method": "initialize",
"params": { "protocolVersion": "2025-06-18",
"capabilities": { "sampling": {} } } }
```

2. Discovery

Client requests information about available capabilities (tools, resources, prompts).

```
// Client → Server
{ "jsonrpc": "2.0", "id": 2, "method": "tools/list" }
```

CONFIDENTIAL
https://www.criterionnetworks.com/

MCP Protocol Lifecycle

(continued)

3. Execution

Client invokes capabilities based on the Host's needs.

```
// Client → Server
{ "jsonrpc": "2.0", "id": 3, "method": "tools/call",
"params": { "name": "weather",
"arguments": { "location": "San Francisco" } } }
```

4. Termination

Connection is gracefully closed when no longer needed.

```
// Client → Server
{ "jsonrpc": "2.0", "id": 4, "method": "shutdown" }
// After response received:
{ "jsonrpc": "2.0", "method": "exit" }
```

CONFIDENTIAL
https://www.criterionnetworks.com/

Tools Capability: Technical Implementation

Protocol Methods & Examples

Tools Protocol Methods

- **tools/list** - Get available tools from the server
- **tools/call** - Invoke a specific tool with arguments

Tool Definition Example

```
{
"name": "get_weather",
"description": "Get current weather for a location",
"inputSchema": {
"type": "object",
"properties": {
"location": {
"type": "string",
"description": "City name"
}
},
"required": ["location"]
}
}
```

CONFIDENTIAL https://www.criterionnetworks.com/

Tools Capability: Technical Implementation

(continued)

CONFIDENTIAL
https://www.criterionnetworks.com/

Tool Invocation Example

```
// Request
{
"jsonrpc": "2.0",
"id": 1,
"method": "tools/call",
"params": {
"name": "get_weather",
"arguments": {
"location": "San Francisco"
}
}
}

// Response
{
"jsonrpc": "2.0",
"id": 1,
"result": {
"content": [{
"type": "text",
"text": "Temperature: 62°F, Partly cloudy"
}]
}
}
```

Resources Capability: Technical Implementation

Protocol Methods & Examples

Resource Protocol Methods

- **resources/list** - Get available resources from server
- **resources/read** - Read a specific resource's content
- **resources/subscribe** - Subscribe to resource updates
- **resources/unsubscribe** - Unsubscribe from updates

Resource Definition Example

```
{
"uri": "file:///project/README.md",
"name": "Project README",
"description": "Main project documentation",
"mimeType": "text/markdown"
}
```

CONFIDENTIAL
https://www.criterionnetworks.com/

Criterion

Resources Capability: Technical Implementation

(continued)

CONFIDENTIAL
https://www.criterionnetworks.com/

Resource Read Example

```
// Request
{
"jsonrpc": "2.0",
"id": 1,
"method": "resources/read",
"params": {
"uri": "file:///project/README.md"
}
}

// Response
{
"jsonrpc": "2.0",
"id": 1,
"result": {
"contents": [{
"uri": "file:///project/README.md",
"mimeType": "text/markdown",
"text": "# Project Overview..."
}]
}
}
```

CONFIDENTIAL
https://www.criterionnetworks.com/

Prompts Capability: Technical Implementation

Protocol Methods & Examples

Prompts Protocol Methods

- **prompts/list** - Get available prompt templates from the server
- **prompts/get** - Retrieve a specific prompt template with arguments

Prompt Definition Example

```
{
"name": "code_review",
"description": "Review code for best practices",
"arguments": [
{
"name": "language",
"description": "Programming language",
"required": true
}
]
}
```

Prompts Capability: Technical Implementation

(continued)

CONFIDENTIAL
https://www.criterionnetworks.com/

Prompt Usage Example

```
// Request
{
"jsonrpc": "2.0",
"id": 1,
"method": "prompts/get",
"params": {
"name": "code_review",
"arguments": { "language": "Python" }
}
}

// Response
{
"jsonrpc": "2.0",
"id": 1,
"result": {
"messages": [{
"role": "user",
"content": {
"type": "text",
"text": "Review this Python code..."
}
}]
}
}
```

Sampling Capability: Technical Implementation

Protocol Methods & Implementation

Sampling Protocol

Allows servers to request LLM completions through the client. The server provides prompts and the client uses its configured LLM to generate responses.

Model Preferences

- **hints** - Suggest preferred models (e.g., "claude-3-5-sonnet")
- **temperature** - Control randomness (0.0-1.0)
- **maxTokens** - Limit response length

CONFIDENTIAL
https://www.criterionnetworks.com/

Sampling Capability: Technical Implementation

(continued)

CONFIDENTIAL
https://www.criterionnetworks.com/

Security Considerations

Sampling requests MUST require human-in-the-loop approval before execution, as they can incur costs and expose sensitive context to the LLM.

Technical Implementation

```
{
"jsonrpc": "2.0",
"id": 1,
"method": "sampling/createMessage",
"params": {
"messages": [{
"role": "user",
"content": {
"type": "text",
"text": "Analyze this data"
}
}],
"modelPreferences": {
"hints": ["claude-3-5-sonnet"],
"temperature": 0.7
},
"maxTokens": 1000
}
}
```

Elicitation Capability: Technical Implementation

Protocol Methods & Examples

Elicitation Protocol

Allows servers to request user input through the client. The server specifies the type of input needed and the client prompts the user accordingly.

Input Types

- **text** - Free-form text input
- **number** - Numeric input
- **boolean** - Yes/no or true/false
- **select** - Choice from predefined options

Request/Response Example

```
// Request
{
"jsonrpc": "2.0",
"id": 1,
"method": "elicitation/request",
"params": {
"prompt": "Enter API key",
"inputType": "text"
}
}

// Response
{
"jsonrpc": "2.0",
"id": 1,
"result": {
"input": "sk-abc123..."
}
}
```

CONFIDENTIAL
https://www.criterionnetworks.com/

MCP Notification System

CONFIDENTIAL
https://www.criterionnetworks.com/

Notification Types

One-way messages sent from server to client without expecting a response. Three main categories:

1. Capability Updates

- **notifications/initialized** - Server ready after initialization
- **notifications/tools/list_changed** - Available tools updated
- **notifications/resources/list_changed** - Resource list changed

2. Resource Updates

- **notifications/resources/updated** - Specific resource content changed
- Includes URI parameter to identify the updated resource

3. Progress Updates

Track long-running operations with progress token, current value, and total.

```
{
"jsonrpc": "2.0",
"method": "notifications/progress",
"params": {
"progressToken": "task-123",
"progress": 50,
"total": 100
}
}
```


Criterion

MCP Security Implementation

Transport Security

- TLS 1.3+ for encrypted communication
- Certificate validation and mutual TLS support

Authentication & Authorization

- OAuth 2.0 / JWT token-based authentication
- Role-based access control (RBAC)
- Capability-based permissions

Content Validation

- Input sanitization and validation
- Output filtering and escaping
- Schema validation for all messages

Human-in-the-Loop Controls

- User approval for sensitive operations (sampling, tool execution)
- Audit logging and monitoring
- Rate limiting and quota management

Implementation Recommendations

- Never trust client-provided data without validation
- Implement defense-in-depth security layers
- Use principle of least privilege for all capabilities
- Regularly audit and rotate credentials
- Monitor for anomalous behavior patterns

CONFIDENTIAL
https://www.criterionnetworks.com/

MCP SDK Implementation

Official SDKs for multiple languages

SDK Architecture & Examples

SDK Architecture

- **TypeScript** - Official reference implementation
- **Python** - Full-featured client & server
- **Rust** - High-performance implementation
- **Go** - Lightweight and efficient
- **Java** - Enterprise-ready
- **C#** - .NET integration

Client SDK Example (Python)

```
# Initialize MCP client
from mcp import Client

client = Client("server-name")
await client.connect()

# Discover and call tools
tools = await client.list_tools()
result = await client.call_tool(
"get_weather",
{"location": "San Francisco"}
)

print(result.content[0].text)
```

CONFIDENTIAL
https://www.criterionnetworks.com/

The **Future** of AI Integration with **MCP**

Universal AI Connectivity

- Connect any AI to any data source
- Eliminate vendor lock-in
- Standardized protocol across platforms

Composable Intelligence

- Mix and match AI capabilities
- Build custom workflows
- Reusable components and tools

Democratized AI Access

- Lower barriers to AI adoption
- Open-source and community-driven
- Accessible to developers of all levels

CONFIDENTIAL
https://www.criterionnetworks.com/

The **Future** of AI Integration with **MCP**

(continued)

"MCP represents a fundamental shift in how AI systems integrate with the world"

— Anthropic, 2024

CONFIDENTIAL
https://www.criterionnetworks.com/

Criterion

THANK YOU

CONFIDENTIAL
https://www.criterionnetworks.com/

Criterion